# Recovering Policy-Induced Errors: Benchmarking and Trajectory Synthesis for Robust GUI Agents

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Paper version: arXiv 2605.29447v1 · CUA Radar Reel poster, not made or endorsed by the authors

## Problem

> *Robustness to policy-induced errors remains under-emphasized in both evaluation and training.*

Agents' **own** mistakes are rarely measured or trained against, in either evals or data.

## Motivation: three open gaps

1. **Coverage.** Data skew to low-level human errors.
2. **Horizon.** Data errors are short; real ones surface late.
3. **Metrics.** Few measure detection and long recovery.

## Evaluation setup

OSWorld-Verified · 369 | WindowsAgentArena · 154 | GUI-RobustEval · 1,216 | Qwen2.5-VL 7B / 32B | 800k SFT samples | 4 runs; 3 on RobustEval

- **All online** (live OS); no offline step accuracy.
- **Held out:** 20k tasks disjoint from evals.

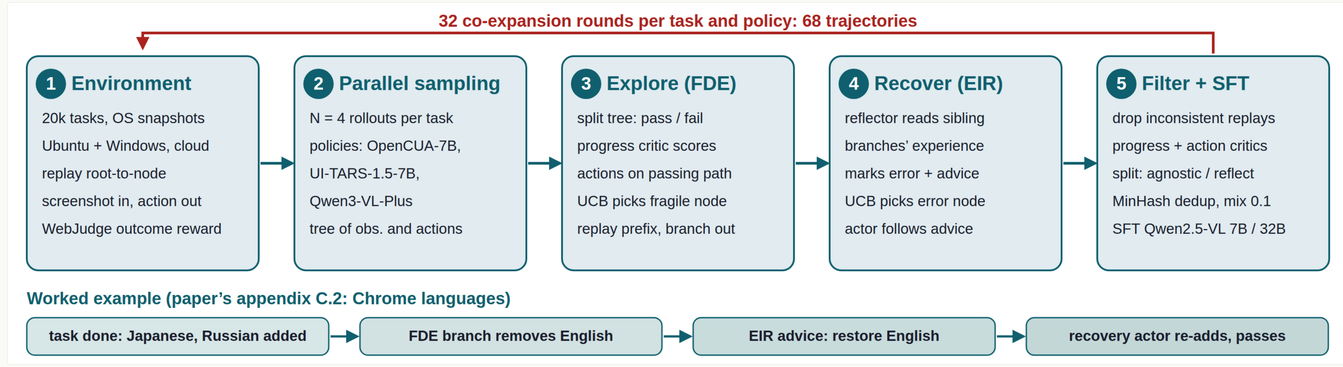
## Our read (CUA Radar)

| 69 | ★★ | online |
|---|---|---|
| score of 100 | band | evidence level |

- **Reports:** online eval; held-out tasks; matched rollout budget; cost; component ablation; judge check; code.
- **Flags:** none recorded. **Not reported:** variance (no std / CI).
- Pattern K26. Authors' numbers, not rerun.

## Method: grow a trajectory tree, keep clean steps, SFT

- **Sees / acts:** screenshot + history in; Thought + PyAutoGUI action out.
- **Tree:** FDE branches from fragile nodes; EIR replays error states with advice.
- **Signal:** WebJudge reward + two critics filter steps; plain SFT, no RL.




$$f_i = (1 - \bar{r}_i) + c\sqrt{\frac{\ln\left(V^f_{p(i)} + 1\right)}{V^f_i + 1}}$$

*fragility score: low step success plus few visits*

## Key results: OSWorld-Verified success rate (%), online

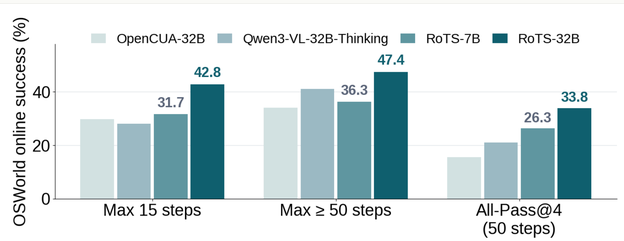



*Table 3, redrawn; RoTS = authors' re-runs.*

**RoTS-32B: 47.4% at 50 steps; 33.8% All-Pass@4**

SO WHAT → *Not charted: Claude 4.5 Sonnet 58.1.*

## Ablation: OSWorld All-Pass@4 (%) by rollout strategy

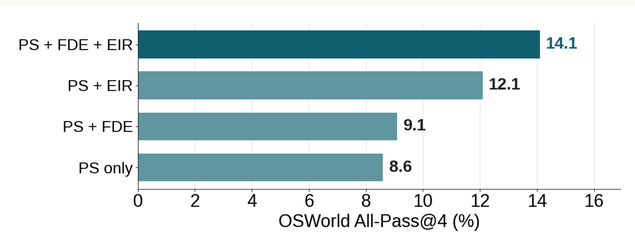



*Table 4, redrawn: 100k SFT on Qwen2.5-VL-7B, same rollout budget, 5k tasks.*

SO WHAT → *EIR lifts All-Pass@4 to 12.1; FDE mostly adds success (18.1 to 19.6). PS = parallel sampling.*

## Headline Numbers

**47.4%**
**RoTS-32B, OSWorld**
*50 steps*

- **33.8%** All-Pass@4 (32B)
- **39.1%** WinAgentArena
- **1,216** RobustEval cases

## The paper's own controls

| CONTROL | ROTS (REPORTED) | COMPARED WITH |
|---|---|---|
| 100k data, OSWorld | RoTS 21.4 / 14.1 | AgentNet 15.3 / 7.8 |
| WindowsAgentArena | 7B 28.2 / 32B 39.1 | Jedi+GPT4-o 32.9 |
| Base, OSWorld 50 st. | RoTS-32B 47.4 | Qwen2.5-VL-32B 3.9 |
| Critic agreement | 88.7 / 90.6 | reward judge 90% |
| Variance | 4 / 3 runs | no std |

## Takeaway

**Synthesized recovery data takes OSWorld to 47.4% (32B, 50 steps); a human-data control trails. Variance not reported.**

800k SFT samples | $48,100 synthesis

## Scan to Read

Paper | Code